

DESIGN AND ANALYSIS OF ALGORITHMS [As per Choice Based Credit System(CBCS)scheme] (Effective from the academic year 2025-2026) SEMESTER – IV			
Course Code	24CS402	CIE Marks	50
Number of Lecture Hours/Week	03	SEE Marks	50
Total Number of Lecture Hours	40	Exam Hours	03
CREDITS - 03			
Course Objectives			
• Explain the mathematical foundation for the analysis of algorithms. • Illustrate the algorithms using brute force and divide and conquer design technique. • Make use of greedy and dynamic algorithmic design techniques for a given problem. • Discuss the problems based on backtracking and branch and bound techniques.			
Module I			Hours
Introduction: Introduction to Algorithm, Fundamentals of Algorithmic Problem Solving. Analysis of Algorithm Efficiency: Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, Mathematical analysis of non- Recursive and recursive Algorithms with Examples. Brute force design techniques: Selection Sort, Bubble Sort, Sequential Search and Brute Force String Matching.			08
Module II			
Decrease-and-Conquer: Topological Sorting, Algorithms for Generating Combinatorial Objects: Generating Permutations, Decrease by-a-Constant- Factor Algorithms: Russian Peasant Multiplication, Variable Size-Decrease Algorithms: Computing Median and the Selection Problem. Depth-First Search and Breadth-First Search Divide and Conquer: General method, Recurrence equation for divide and conquer, solving it using Master's theorem, Merge sort, Quick sort, Binary search			08
Module III			
Transform-and-Conquer: Presorting, Heaps and Heap sort, Horner's Rule. Space and Time Tradeoffs: Horspool's Algorithm, Boyer-Moore Algorithm. Dynamic Programming: Coin Change Problem Transitive Closure: Warshall's Algorithm. All Pairs Shortest Paths: Floyd's Algorithm			08
Module IV			
Greedy Method: Minimum cost spanning trees: Prim's Algorithm, Kruskal's Algorithm with performance analysis.			

Single source shortest paths: Dijkstra's Algorithm. Optimal Tree problem: Huffman Trees and Codes, Fractional Knapsack Problem	08
Module V	
Backtracking: General method ,n-Queens Problem, Subset-Sum Problem. Branch-and-Bound: Assignment Problem ,Knapsack Problem, Traveling Salesman Problem. NP-Completeness: Polynomial time, Polynomial-time verification, NP- completeness and reducibility. NP-Complete Problems: The Clique problem, The Vertex Cover problem.	08

Text Books:

- 1.Introduction to the Design and Analysis of Algorithms, Anany Levitin:, 2nd Edition, 2009.Pearson.
- 2.Computer Algorithms/C++, Ellis Horowitz, Satraj Sahni and Rajasekaran, 2nd Edition, 2014,Universities Press/

Reference Books:

- 1.Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
- 2.Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education).

Question paper pattern:

- The SEE question paper will be set for 100 marks and the marks scored will be proportionately reduced to 50.
- The question paper will have ten full questions carrying equal marks.
- Each full question carries 20 marks.
- There will be two full questions (with a maximum of three sub questions) from each module
- Each full question will have sub questions covering all the topics under a module.
- The students will have to answer five full questions, selecting one full question from each module